BEFORE YOU READ

Fourteen limits in five families

The special limits on this page group by the functions involved and by where x is heading.

L0 **Special limits**

L1 **Trigonometric, $x \to 0$**

L2 **$\sin x / x$**	1
L2 **$(1 - \cos x) / x$**	0
L2 **$(1 - \cos x) / x^2$**	1/2
L2 **$\tan x / x$**	1
L2 **$x / \sin x$**	1

L1 **Exponential, $x \to 0$**

L2 **$(e^x - 1) / x$**	1
L2 **$(a^x - 1) / x$** $a > 0$	$\ln a$
L2 **$\ln(1 + x) / x$**	1

L1 **Definition of e**

L2 **$(1 + 1/x)^x$ as $x \to \infty$** equivalently $(1 + x)^{1/x}$ as $x \to 0$	e

L1 **Logarithmic boundary, $x \to 0^+$**

L2 **$x \ln x$**	0
L2 **$x^n \ln x$** $n > 0$	0

L1 **Growth rate, $x \to \infty$**

L2 **$\ln x / x^n$** $n > 0$	0
L2 **x^n / e^x** any n	0
L2 **x^n / a^x** any n, $a > 1$	0

TAKEAWAY

Two directions

The trigonometric and exponential limits look at $x \to 0$; the growth-rate limits look at $x \to \infty$ and rank logarithms below powers below exponentials.